

# A Practical Guide to Hidden Markov model

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TOPIC -- HIDDEN MARKOV MODEL

$$P(O | \lambda) = \sum_Q P(O | Q, \lambda) P(Q | \lambda)$$

-- 01 --

In its discrete form, a hidden Markov process can be visualized as a generalization of the urn problem with replacement (where each item from the urn is returned to the original urn before the next step).

-- 02 --

In this piece of code, `start_probability` represents Alice's belief about which state the HMM is in when Bob first calls her (all she knows is that it tends to be rainy on average). The particular probability distribution used here is not the equilibrium one, which is (given the transition probabilities) approximately {'Rainy': 0.57, 'Sunny': 0.43}. The `transition_probability` represents the change of the weather in the underlying Markov chain. In this example, there is only a 30% chance that tomorrow will be sunny if today is rainy. The `emission_probability` represents how likely Bob is to perform a certain activity on each day. If it is rainy, there is a 50% chance that he is cleaning his apartment; if it is sunny, there is a 60% chance that he is outside for a walk.

-- 03 --

**Probability of an observed sequence** The task is to compute in a best way, given the parameters of the model, the probability of a particular output sequence. This requires summation over all possible state sequences: The probability of observing a sequence

-- 04 --

A hidden Markov model (HMM) is a Markov model in which the observations are dependent on a latent (or hidden) Markov process (referred to as  $X$ ). An HMM requires that there be an observable process  $Y$  whose outcomes depend on the outcomes of  $X$  in a known way. Since  $X$  cannot be observed directly, the goal is to learn about state of  $X$  by observing  $Y$ . By definition of

being a Markov model, an HMM has an additional requirement that the outcome of  $Y$  at time  $t = t_0$  must be "influenced" exclusively by the outcome of  $X$  at  $t = t_0$  and that the outcomes of  $X$  and  $Y$  at  $t < t_0$  must be conditionally independent of  $Y$  at  $t = t_0$  given  $X$  at time  $t = t_0$ . Estimation of the parameters in an HMM can be performed using maximum likelihood estimation. For linear chain HMMs, the Baum–Welch algorithm can be used to estimate parameters. Hidden Markov models are known for their applications to thermodynamics, statistical mechanics, physics, chemistry, economics, finance, signal processing, information theory, pattern recognition—such as speech recognition, handwriting recognition, gesture recognition, part-of-speech tagging, musical score following, partial discharges and bioinformatics.

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$$P \left( Y_{t_0} \in A \mid \{X_t \in B_t\}_{t \leq t_0} \right) = P \left( Y_{t_0} \in A \mid X_{t_0} \in B_{t_0} \right)$$
  

$$\{ \text{operatorname} \{ \mathbf{P} \} \} (Y_{t_0} \in A \mid \{X_t \in B_t\}_{t \leq t_0}) = \text{operatorname} \{ \mathbf{P} \} (Y_{t_0} \in A \mid X_{t_0} \in B_{t_0})$$
, for every  $t_0$ , every Borel set  $A$ , and every family of Borel sets  $\{B_t\}_{t \leq t_0}$ .

-- 06 --

$$P \left( Y_n \in A \mid X_1 = x_1, \dots, X_n = x_n \right) = P \left( Y_n \in A \mid X_n = x_n \right)$$
  

$$\{ \text{operatorname} \{ \mathbf{P} \} \} \{ \text{bigl} \{ Y_n \in A \mid X_1 = x_1, \dots, X_n = x_n \} \bigr \} = \text{operatorname} \{ \mathbf{P} \} \{ \text{bigl} \{ Y_n \in A \mid X_n = x_n \} \bigr \}$$
, for every  $n \geq 1$ ,  $x_1, \dots, x_n$ , and every Borel set  $A$ . Let  $X_t$  and  $Y_t$  be continuous-time stochastic processes. The pair  $(X_t, Y_t)$  is a hidden Markov model if

-- 07 --

Concepts Teif, V. B.; Rippe, K. (2010). "Statistical–mechanical lattice models for protein–DNA binding in chromatin". *J. Phys.: Condens. Matter.* 22 (41) 414105. arXiv:1004.5514. Bibcode:2010JPCM...22O4105T. doi:10.1088/0953-8984/22/41/414105. PMID 21386588. S2CID 103345. A Revealing Introduction to Hidden Markov Models by Mark Stamp, San Jose State University. Fitting HMM's with expectation-maximization – complete derivation A step-by-step tutorial on HMMs Archived 2017-08-13 at the Wayback

Machine (University of Leeds) Hidden Markov Models (an exposition using basic mathematics) Hidden Markov Models (by Narada Warakagoda) Hidden Markov Models: Fundamentals and Applications Part 1, Part 2 (by V. Petrushin) Lecture on a Spreadsheet by Jason Eisner, Video and interactive spreadsheet

-- 08 --

The choice of the urn for the  $n$ -th ball depends only upon a random number and the choice of the urn for the  $(n - 1)$ -th ball. The choice of urn does not directly depend on the urns chosen before this single previous urn. Therefore, this is called a Markov process. It can be described by the upper part of Figure 1.

-- 09 --

Bayesian modeling of the transitions probabilities Hidden Markov models are generative models, in which the joint distribution of observations and hidden states, or equivalently both the prior distribution of hidden states (the transition probabilities) and conditional distribution of observations given states (the emission probabilities), is modeled. The above algorithms implicitly assume a uniform prior distribution over the transition probabilities. However, it is also possible to create hidden Markov models with other types of prior distributions. An obvious candidate, given the categorical distribution of the transition probabilities, is the Dirichlet distribution, which is the conjugate prior distribution of the categorical distribution. Typically, a symmetric Dirichlet distribution is chosen, reflecting ignorance about which states are inherently more likely than others. The single parameter of this distribution (termed the concentration parameter) controls the relative density or sparseness of the resulting transition matrix. A choice of 1 yields a uniform distribution. Values greater than 1 produce a dense matrix, in which the transition probabilities between pairs of states are likely to be nearly equal. Values less than 1 result in a sparse matrix in which, for each given source state, only a small number of destination states have non-negligible transition probabilities. It is also possible to use a two-level prior Dirichlet distribution, in which one Dirichlet distribution (the upper distribution) governs the parameters of another Dirichlet distribution (the lower distribution), which in turn governs the transition probabilities. The upper distribution governs the overall distribution of states, determining how likely each state is to occur; its concentration parameter determines the density or sparseness of states. Such a two-level prior distribution, where both concentration parameters are set to produce sparse distributions, might be useful for example in unsupervised part-of-speech tagging, where some parts of speech occur much more commonly than others; learning algorithms that assume a uniform prior distribution generally perform poorly on this task. The parameters of models of

this sort, with non-uniform prior distributions, can be learned using Gibbs sampling or extended versions of the expectation-maximization algorithm. An extension of the previously described hidden Markov models with Dirichlet priors uses a Dirichlet process in place of a Dirichlet distribution. This type of model allows for an unknown and potentially infinite number of states. It is common to use a two-level Dirichlet process, similar to the previously described model with two levels of Dirichlet distributions. Such a model is called a hierarchical Dirichlet process hidden Markov model, or HDP-HMM for short. It was originally described under the name "Infinite Hidden Markov Model" and was further formalized in "Hierarchical Dirichlet Processes".